Econ 5015 Fall 2026 Data Wrangling Assignment

You may work alone or in groups of 2. If in a group, submit one script for the group with both names listed via Gradescope.

Name your script **crimedata.py**

You should import pandas.

(A) The files http://ashafran.github.io/crime_*.csv contain data on total crimes and crime rates for 51 states over time (one file for each state plus DC) downloaded from the FBI Uniform Crime Reports where the * is number ranging from 1 to 51 (inclusive). You should import the data into a single python data frame called **full_data** with one row for each state/year. I recommend using pd.concat, as shown in the Pandas/Dataframe notebook from earlier this term. You should use a loop to automate the reading of the data – do NOT have 51 lines of code to read the data.

(B) The dataframe from (A) has one column for population and 9 columns that are the *total* values for 9 classes of crimes (and also other columns with crime rates that we will ignore for this part). Create a new data frame called **us_data** that has one observation per year and has 10 columns. The index of the data frame should be the year. The 10 columns should correspond to columns for population and the 9 crime *total* columns. These 10 columns should be set equal to the sum of all observations in that year across all states. For example, the population columns should be the total U.S. population in each year, and the other columns should be the total number of crimes of each type in the U.S. in each year. I recommend using groupby for this part.